

Summary

- Basic Facts About Financial Structure:
 - We reviewed 8 basic facts concerning the structure of the financial system
- Transaction Costs:
 - How transaction costs can hinder capital flow.
 - The role financial institutions play in reducing transaction costs

Summary

- Asymmetric Information leads to
 1. Adverse Selection.
 2. Moral Hazard
- How Adverse Selection Influences Financial Structure
 - How adverse selection effects the flow of capital.
 - Who can reduce this problem?
 1. Rating agencies
 2. Government Regulation to Increase Information
 3. Financial Intermediation
- The free-rider problem
 - If some use information for free.
 - Explains why banks play a more important role in financing the than securities

How Moral Hazard Affects the Choice Between Debt and Equity Contracts



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Tools to Help Solve the Principal-Agent Problem

1. Production of Information: Monitoring
 2. Government Regulation to Increase Information
 3. Financial Intermediation (e.g., venture capital)
- } Equity Financing

4. Debt Contracts (Debt Financing)

- Most debt contracts require the borrower to pay a fixed amount (interest) and borrower keep any profit above this amount
- Only cares about fixed return (interests+principle).
- Equity holder vs. Debt holder
 - Lower cost of state verification.

→ More willing to become a debt holder if do not have enough information or low risk preference.

How Moral Hazard Influences Financial Structure in Debt Markets

- Do you think debt could be still subject to moral hazard?
- For example, what if a firm owes \$100 in interest, but only has \$90? It is essentially a bankrupt
- What you will do, if you're the firms' manager?
 - More risk taking or risk aversion?
- Who suffers from moral hazard?
 - Firm or lender?

How Moral Hazard Influences Financial Structure in Debt Markets

- When you decide to lend, what could go wrong? (What moral hazard you may suffer?)
 - Risk preference is different between you and borrower
- Borrowers have an incentive to take on investment projects that are riskier than the lenders would like. Why?.
 - If succeed in the project, who benefit?
 - If lost, who suffer?

How Moral Hazard Influences Financial Structure in Debt Markets

- Tools to Help Solve Moral Hazard in Debt Contracts
 1. Net Worth is high or Collateral.
 - Net Worth = Assets - Liabilities
 - Owner put 90,000, instead of 10,000 in the firm
 - Borrowers have more to lose, compared with low net worth or no collateral.
- Reduced risk in the investment of borrowers.
- E.g. You should choose borrower that

How Moral Hazard Influences Financial Structure in Debt Markets

- Tools to Help Solve Moral Hazard in Debt Contracts

1. Net Worth is high or Collateral

2. Monitoring and Enforcement of Restrictive Covenants.

Examples are covenants that ...

1. Discourage undesirable behavior

- Borrowing only used for specific activities.
 - Eg. the loan can only be used to buy particular equipment.

2. Encourage desirable behavior

- Must maintain minimum holders of certain amount assets.
- Eg. Must buy life insurance for mortgage

How Moral Hazard Influences Financial Structure in Debt Markets

- Tools to Help Solve Moral Hazard in Debt Contracts
 1. Net Worth is high or Collateral
 2. Monitoring and Enforcement of Restrictive Covenants.
Examples are covenants that ...
 1. Discourage undesirable behavior
 2. Encourage desirable behavior
 3. Keep collateral valuable
 - E.g. good condition of your pledged car.
 4. Provide information
 - Report about its activities periodically
 - Lenders have right to audit
- Hence, we see complicated debt contract (Fact #8).

How Moral Hazard Influences Financial Structure in Debt Markets

- Tools to Help Solve Moral Hazard in Debt Contracts
 1. Net Worth is high or Collateral
 2. Monitoring and Enforcement of Restrictive Covenants.
- What are the problems?
 - Is it possible to write covenants that rule out every risks?.
 - Monitoring and enforcement of restrictive covenants are costly.
 - Meaningless if the borrower knows that the lender won't check up or enforcement is difficult

How Moral Hazard Influences Financial Structure in Debt Markets

- Tools to Help Solve Moral Hazard in Debt Contracts
 1. Net Worth is high or Collateral
 2. Monitoring and Enforcement of Restrictive Covenants.
- What are the problems?
 - Cannot rule out every risks
 - Monitoring and enforcement are costly
 - Free-rider problem for open-traded debt
 - Eg. Corporate bonds, many other debt holders
 - Everyone wants to be a free rider.
 - No one is actively monitoring firms.
 - → Moral hazard

How Moral Hazard Influences Financial Structure in Debt Markets

- How to Solve Moral Hazard in Debt Contracts?
 1. Net Worth is high or Collateral
 2. Monitoring and Enforcement of Restrictive Covenants.
- What are the problems?
 - Cannot rule out every risks
 - Monitoring and enforcement are costly
 - Free-rider problem for open-traded debt

Summary: How Moral Hazard Influences the Choice Between Debt and Equity Contracts

- Tools to Help Solve asymmetric information
- For equity holders

(e.g. Principal-Agent Problem)

1. Production of Information: Monitoring
2. Government Regulation to Increase Information
3. Financial Intermediation (e.g., venture capital)
4. Choose to become debt holders

For debt holders

Financial Intermediation. ☺

- Private loans → Free-rider decreases
- Explains Facts # 3–4

Summary: How Moral Hazard Influences Financial Structure in Debt Markets

- 4. Debt could be still subject to moral hazard
- Tools to Help Solve Moral Hazard in Debt Contracts
 1. Net Worth is high or Collateral
 2. Monitoring and Enforcement of Restrictive Covenants

Recall: Asymmetric Information Problems

- Adverse selection and moral hazard problems
- **Free-rider problem** for public traded securities.
- Financial intermediaries, especially banks, make private securities.
→ Play a greater role in financing business.

Table 1 Summary of Asymmetric Information Problems and Tools to Solve Them (1 of 3)

Asymmetric Information Problem	Tools to Solve It	Explains Fact Number
	Private production and sale of information. i.e. Rating agency	1,2
	Government regulation to increase information	5
	Financial intermediation	3,4,6
	Collateral and net worth	7

1. Stocks are not the most important source of external financing.
2. Marketable securities are not the primary source of finance.
3. Indirect finance is more important than direct finance.
4. Banks are the most important source of external funds.
5. The financial system is heavily regulated.
6. Only large, well-established firms have access to securities markets.
7. Collateral is prevalent in debt contracts.
8. Debt contracts have numerous restrictive covenants.

Table 1 Summary of Asymmetric Information Problems and Tools to Solve Them (2 of 3)

Asymmetric Information Problem	Tools to Solve It	Explains Fact Number
 leads to <u>Moral hazard</u> in equity financing (e.g. principal-agent problem)	Production of information monitoring.	1
	Government regulation to increase information	5
	Financial intermediation	3
	Debt contracts	6,7

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Table 1 Summary of Asymmetric Information Problems and Tools to Solve Them (3 of 3)

Asymmetric Information Problem	Tools to Solve It	Explains Fact Number
 Moral hazard in debt financing	Collateral and net worth	6,7
	Monitoring and enforcement of restrictive covenants	8
	Financial intermediation	3,4

1. Stocks are not the most important source of external financing.
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Financial Development and Economic Growth

- Importance of Financial intermediaries to
 - Firms and investors
 - How about economic growth of a country?



Financial Development and Economic Growth

- Financial underdevelopment leads to low growth
- Why underdeveloped financial systems?
 1. Weak accounting standards.
 2. Government directs credit (state-owned banks)
 3. Inadequate government regulation
 4. Poor legal system
- **Property rights without enforcement are meaningless.**
 - Each society regulates, by a combination of formal legislation, common law and custom, what counts as an item of property, and how that item may be used.

Importance of efficient legal system

- Eg. The Tyranny of Collateral
- Legal title to land in some developing countries is complicated
 - Legal title to a dwelling land in the Philippines involves 168 bureaucratic steps
 - In Haiti? 176 steps over nineteen years
- Trillions is held in land not legally titled. And in that case, land cannot be used as collateral for loans.
- Adverse selection problem becomes worse

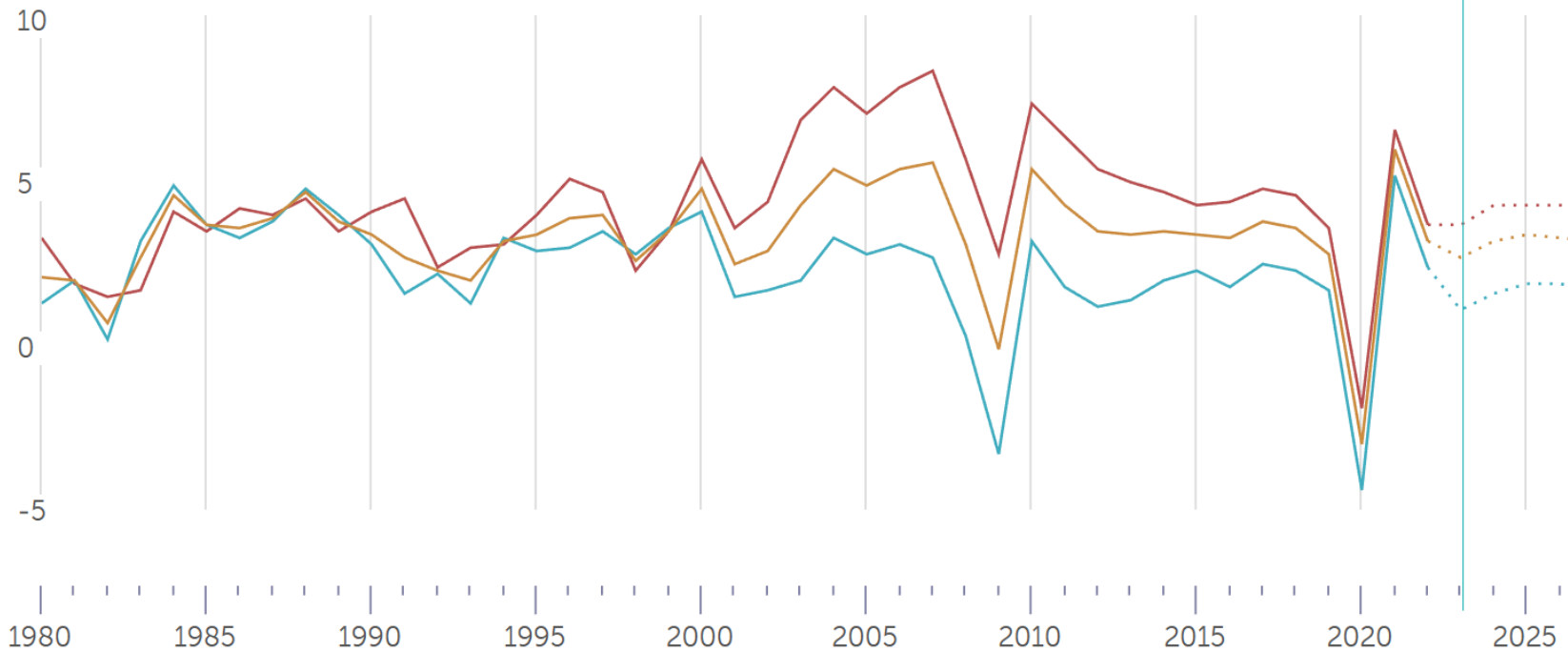
Is China a Counter-example? (1 of 2)

- Even with its booming economy, China's financial development is still in an early stage.

Economic development globally

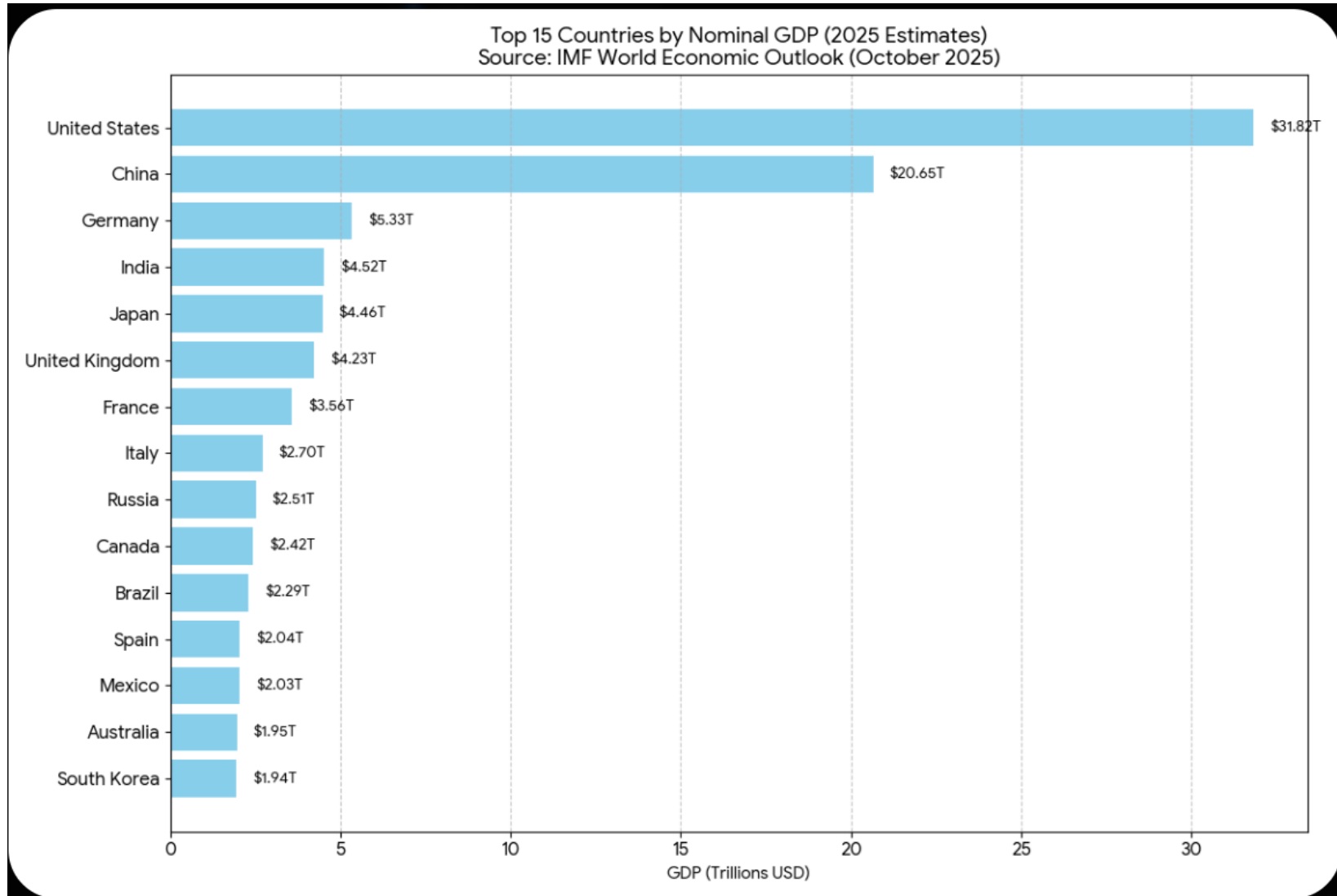
TREND (1980-2027)

Annual percent change in GDP

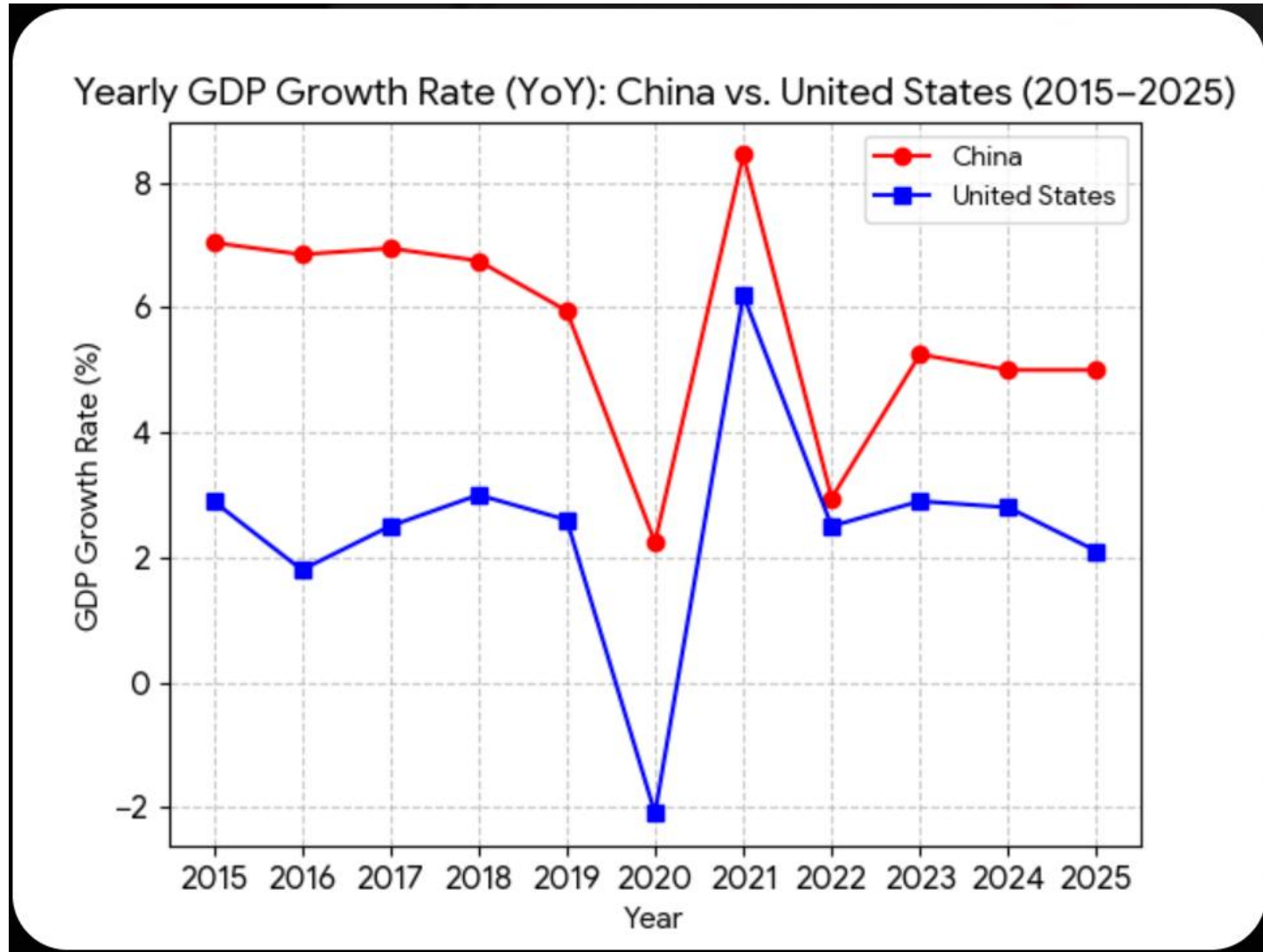


- Emerging market and developing economies
- Advanced economies
- World

Top Countries by Nominal GDP (2025 Estimates)



2020-2025 GDP growth rate



Is China a Counter-example? (1 of 2)

- Even with its booming economy, China's financial development is still in an early stage.
- Why booming?
 - High growth fueled by a high savings rate, a massive buildup of capital
 - Shift a massive pool of underutilized labor from the subsistence-agriculture sector into higher-productivity activities

Is China a Counter-example? (2 of 2)

- Financial market in China
 - Improve the efficiency of capital allocation (Equity financing)
 - Approval System → Registration system
 - Especially banking system (Debt financing)
 - State-owned banks → Privatization or mixed ownership
 - Commercial banks engage in investment banking business
 - Legal reform
 - Eg. New bankruptcy law



Financial Crises and Aggregate Economic Activity



Our analysis of the effects of **adverse selection** and **moral hazard** can also assist us in understanding **financial crises**.

The end result of most financial crises is the inability of markets to channel funds from savers to productive investment opportunities.

Financial Intermediaries: Every coin has two sides

——Conflicts of Interest

- **Bright side** of financial intermediaries
- A cost advantage :
 - Expertise in interpreting signals and collecting information
 - Develop broader and longer-term relationships with firms
- Economies of scale:
 - Use the information over and over again, providing multiple financial services to different customers (e.g. loans and bonds)
 - Develop broader and longer-term relationships with firms.

Financial Intermediaries: Every cion has two sides

——Conflicts of Interest

- Impact of financial intermediaries societies



Financial Intermediaries: Every cion has two sides

——Conflicts of Interest

- By serving different customers
 - Bright side of financial intermediaries
 - **Dark side**—— Conflicts of interest
 - Customer A's interest conflicts with customer B.
- Conflicts of interest
 - A type of moral hazard that occurs when a person or institution has multiple interests, and serving one interest is detrimental to the other
 - Occurs when having multiple objectives (interests).

Why we care about conflicts of interest?

- Corporate and accounting scandals due to conflicts of interest have received tremendous public attention
 - Eg. Rating agency: they rate and provide advise
 - Investors loss millions of dollar because of their bad rating
- FI talk up about firms due to conflicts of interest
- Bankruptcies have cost employees of these firms their jobs and their pensions, and because the scandals may have hampered the efficient functioning of the financial system.



Conflicts of Interest

- What do Conflicts of interest lead to?
 1. Reduction in the quality of information in financial markets→
 2. Increases asymmetric information problems→
 3. Prevents financial markets from channeling funds into the most productive investment opportunities →
 4. The financial markets and the economy become less efficient.
- 3 classic conflicts
 - Looking at these closely offers insight in avoiding these conflicts in the future.

Conflicts of Interest: Investment Banking

(1 of 3)

- Investment banks performance TWO tasks:
 1. Research companies with public securities
 2. Underwrite securities for companies for sale to the public.
- Research is expected to be unbiased and accurate, reflecting the facts about the firm. It is used by the **public** to form investment choices.
- Underwriters can better serve the **firm** going public if the firm's outlook is optimistic.
 - Underwriters will have an easier time if research is positive.
- What is the bright side of having both services?
- What are the conflicts of interest?

Conflicts of Interest: Underwriting and Research in Investment Banking (2 of 3)

- The interest of the issuing firm
 - optimistic research
- The interest of the public
 - unbiased research
- Investment banks serve the interest of the issuing **firm** or the **public**?



- “Our objective . . . is to adopt a policy, fully understood by the entire firm, including the Research Department, that we do not make negative or controversial comments about our clients as a matter of sound business practice.”

——an internal Morgan Stanley memo

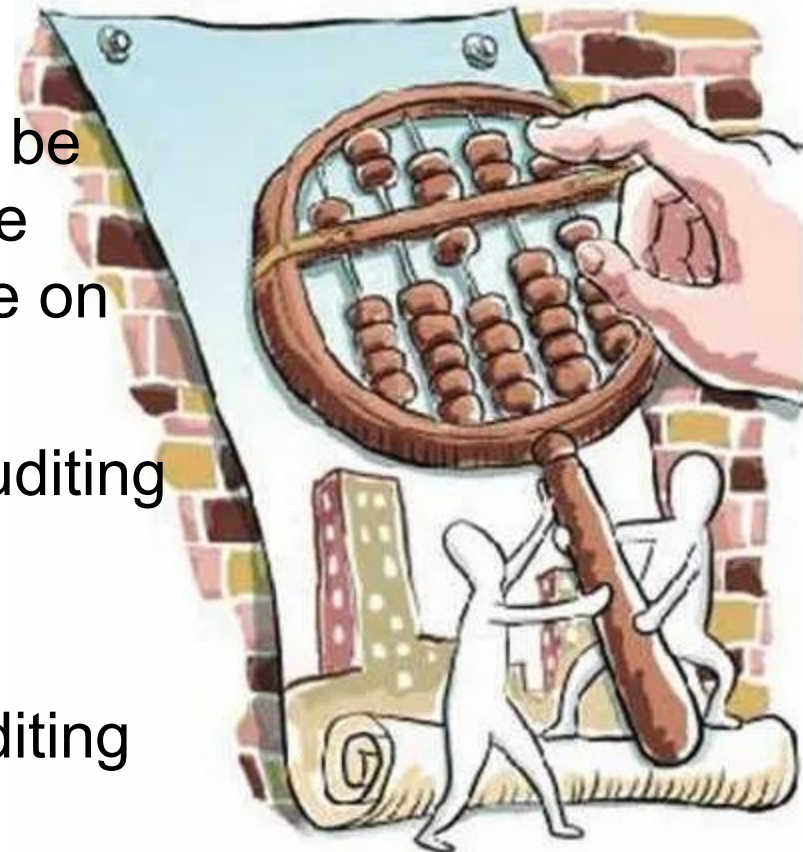
Conflicts of Interest: Accounting Firms

(1 of 2)

- TWO services in auditing firms
 1. Auditing service
 2. Consulting service
- Auditors **check** the assets and books of a firm for the quality and accuracy of the information.
 - an unbiased opinion of the firm's financial health.
- **Consultants**, for a fee, help firms with variety of managerial, strategic, and operational projects.
- What is the bright side of having both services?
- What are the conflicts of interest?

Conflicts of Interest: Auditing and Consulting in Accounting Firms (2 of 2)

- Two potential sources of conflicts of interest:
 - Consulting service is very profitable. And if you provide negative audit opinion on this client...
 - Firms tax or financial plans may be designed by the consultant in the auditing firms. And if you criticize on these...
- Interest of public and client in the auditing service.
 - Public: Unbiased audit opinion
 - Firms be audited: Favorable auditing opinion.



Conflicts of Interest: Rating Agencies (1 of 3)

- What is a credit rating agency?
 - assign a credit rating to a security issuance of a firm based on projected cash flow, assets pledged, etc. The rating helps determine the riskiness of a security
- Conflicts of interest:

Act in the interests of **corporate clients** vs. **public investors**
- TWO potential sources of conflicts of interest:
 1. The issuers of securities pay a rating firm such as S&P's or Moody's to have their securities rated.
 2. Consultants, for a fee, help firms with variety of managerial, strategic, and operational projects.
 - An rating agency acting as both a rater and consultant for a firm clearly is not objective, especially if the consulting fees exceed the rating fees.

Conflicts of Interest: Credit Assessment and Consulting in Rating Agencies

(2 of 3)

Rating agencies, such as Moody's and Standard and Poor, were caught in this game during the housing bubble

Firms asked the rater to help structure debt offerings to attain the highest rating possible

When the debt subsequently defaulted, it was difficult for the agency to justify the original high rating. Perhaps it was just error.

But few believe that—most see the rating agencies as being blinded by high consulting fees.

Summary:

3 classic Conflicts of Interest

1. Underwriting and Research in Investment Banking
2. Auditing and Consulting in Accounting Firms
3. Credit Assessment and Consulting in Rating Agencies

If you were regulators, what could be done to mitigate the conflicts of interest?

Remedies? (1 of 3)

- Two major policy measures.
 1. Sarbanes-Oxley Act of 2002
 - Established an oversight board to supervise accounting firms
 - Increased the SEC's budget for supervisory activities
 - Limited consulting relationships between auditors and firms
 - Enhanced criminal charges for obstruction
 - Improved the quality of the financial statements and board
- Reduced conflicts of interest

Remedies? (2 of 3)

2. Global Legal Settlement of 2002.

- Required investment banks to sever links between research and underwriting.
 - Spinning is explicitly banned
 - Imposed a \$1.4 billion fine on accused investment banks
 - Added additional requirements to ensure independence and objectivity of research reports
- Reduced conflicts of interest

Remedies? (3 of 3)

- Do these work?
 - There is much criticism over the cost involved with these separations. In other words, financial institutions can no longer take advantage of the economies of scope gained from relationships.
 - Some have argued that Sarbanes-Oxley has negatively impacted the economy.
 - Regulatory costs are too high.

- Can we



Chapter Summary (3 of 4)

- How Moral Hazard Affects the Choice Between Debt and Equity Contracts:
 - Principal-agent problem.
 - How moral hazard influences the use of more debt than equity
- How Moral Hazard Influences Financial Structure in Debt Markets:
 - How moral hazard and debt may lead to increased risk-taking
 - Tools to reduce this problem
 - Financial intermediaries

Chapter Summary (4 of 4)

- Conflicts of Interest.
- Three types of financial service activities
- Two major policy measures